

WHEN DOES JOINT KEY-CACHE ALLOCATION PAY? CONTEXT-DEPENDENT REGIMES IN QUANTIZATION AND EVICTION

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ABSTRACT

KV-cache compressors can spend a memory budget by quantizing every token, evicting tokens, or mixing both actions. Existing work shows how to build such hybrids; we ask when the mixed interior improves over its two endpoints. Under a separable local surrogate for attention-output distortion, key-token precision is a discrete rate-allocation problem that includes a normalized zero-rate cost. A finite-bit tier is dominated when its absolute logit-noise cost exceeds that zero-rate cost, yielding a measurable tier-extinction coordinate. Across six LLMs and 25 model-context cells spanning 4k–256k, with allocations evaluated by exact attention-output recomputation, this coordinate orders the mixed allocator’s advantage under a symmetric lagged-information comparison (Spearman $\rho = -0.978, -0.915$ after accounting for model identity). The crossing is architecture-conditioned rather than universal, and increasing context moves fixed models toward an eviction-like corner. Measurements over 4,096 decode tokens further reveal two output-error timescales: the head-level family label remains stable (p90 regret 1.00 in six cells), while a frozen token allocation costs $2.52\text{--}4.30\times$ a one-step-old allocation. A completed end-task campaign does not validate the regime map as an accuracy router: uniform quantization is the strongest fixed policy in 34/36 valid cells, and the calibrated router has 17 clear losses. Thus the map characterizes local distortion regimes rather than establishing downstream effectiveness.

1 INTRODUCTION

At long context, the KV cache can bound batch size and decode throughput. A cache budget admits at least three actions: assign every token one low precision (Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2025), retain selected tokens at a higher precision (Zhang et al., 2023; Liu et al., 2023; Li et al., 2024; Xiao et al., 2024), or mix zero and finite rates across tokens. The last option is no longer hypothetical: DiffKV and HqeKV combine pruning with multiple precision levels (Zhang et al., 2025; Wang et al., 2026). What remains unclear is whether the extra interior choices are useful for a given model and deployment context, or whether a simpler endpoint already suffices.

We study that missing question. The motivating reversal is within one model, not between leaderboards: from 8k to 128k, Llama-3.3-70B’s dead-2-bit fraction rises from 5.5% to 41.4%, its hindsight portfolio gain falls from $3.17\times$ to $1.40\times$, and the share of heads where the interior beats both endpoints by $2\times$ falls from 87.4% to 21.3%. A verdict at one context therefore need not transfer to another.

Our lens is a joint key-token action space. We derive a first-order, separable attention-output surrogate for finite-bit key noise; we place eviction at zero rate using an explicitly stated local approximation; and we solve the actual discrete allocation over hardware tiers. This view exposes two different boundaries. Insufficient variation in token importance removes the benefit of unequal precision, while excessive absolute noise makes low-bit tiers worse than zero rate. We call the resulting picture an *architecture-conditioned regime map*; the present data establish its mixed-to-eviction side most strongly.

Our contributions are:

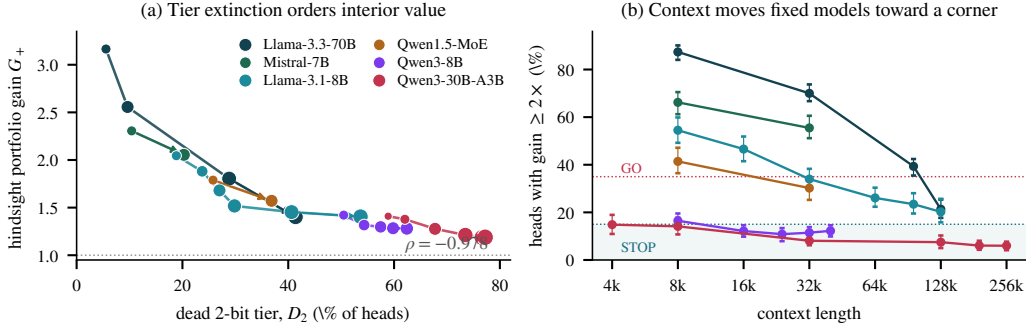


Figure 1: **Measured symmetric results.** Both the mixed allocator and the eviction endpoint use lagged information; exact recomputation scores both. (a) The tier-extinction coordinate D_2 strongly orders the primary continuous response, hindsight portfolio gain G_+ . Arrows show increasing context. (b) The secondary $2\times$ band statistic shows that context moves fixed models across predeclared output-error bands. Error bars are 90% layer-cluster intervals. The 25-cell extension spans 4k–256k; thresholds are not universal across architectures.

1. **A common action space and a regime question.** We formulate within-head key-token precision and eviction under one local output-distortion surrogate, distinguish the exact set-eviction error from the separable zero-rate approximation, and ask when the discrete interior beats both endpoints.
2. **An architecture-conditioned regime indicator.** Across 25 cells, the dead-tier fraction strongly orders the interior’s measured attention-output advantage, including under symmetric lagged information and model-controlled analyses; the critical value remains architecture-specific.
3. **Context traversal and two refresh timescales.** Context changes tier viability within a fixed model. The output-error-optimal family label changes slowly, whereas token allocations go stale quickly; GQA grouping and a key-only score cascade quantify the costs of a future controller.

Concurrent RDKV also derives joint eviction–quantization rate allocation (Zhang et al., 2026). Our paper-specific contribution is complementary: measuring and ordering *when* a nontrivial token-precision interior lowers attention-output error and how that answer changes with context. Section 6 gives a precise comparison to RDKV and other allocation axes.

2 JOINT TOKEN-RATE ALLOCATION AS A COMMON FOUNDATION

Scope and local quantization surrogate. For one query q , keys k_i , and values v_i , let $s_i = q^\top k_i / \sqrt{d}$, $a = \text{softmax}(s)$, and $o = \sum_i a_i v_i$. Quantizing key i perturbs its logit by δ_i . The first-order output change is

$$\Delta o \approx \sum_i a_i (v_i - o) \delta_i. \quad (1)$$

If residual perturbations are zero mean and cross-token covariance terms vanish,

$$\mathbb{E} \|\Delta o\|^2 \approx \sum_i w_i d_{b_i}, \quad w_i = a_i^2 \|v_i - o\|^2, \quad d_b = \text{Var}(\delta_b). \quad (2)$$

Only a common-mode logit shift is canceled exactly by softmax; arbitrary token-dependent bias and covariance are outside this surrogate. We therefore use Eq. 2 to *choose* allocations, never to score their reported error.

Exact eviction versus a zero-rate surrogate. For an evicted set E , define $A_E = \sum_{i \in E} a_i$. Renormalizing the surviving attention gives the exact identity

$$o_{\setminus E} - o = -\frac{\sum_{i \in E} a_i (v_i - o)}{1 - A_E}, \quad \|o_{\setminus E} - o\|^2 = \frac{\|\sum_{i \in E} a_i (v_i - o)\|^2}{(1 - A_E)^2}. \quad (3)$$

The denominator and cross-token inner products make set eviction nonseparable. For a singleton j , the exact error magnitude is $a_j \|v_j - o\| / (1 - a_j)$, as derived by CAOTE (Goel et al., 2025). To include zero rate in a token-separable allocator, we use the small-evicted-mass, cross-term-free approximation $\tilde{D}_0(E) = \sum_{i \in E} w_i$ and normalize its *absolute* cost to $d_0 = 1$. Eviction is therefore a zero-rate action *inside the surrogate*, not an exact independent-token equality.

The implemented discrete allocation. For available integer tiers $\mathcal{B} = \{0, 1, 2, 3, 4, 5, 6, 8\}$ and an average key budget B , we solve

$$\min_{b_i \in \mathcal{B}} \sum_i w_i d_{b_i} \quad \text{s.t.} \quad \sum_i b_i \leq BL, \quad b_i^*(\lambda) = \arg \min_{b \in \mathcal{B}} \{w_i d_b + \lambda b\}. \quad (4)$$

A bisection on λ enforces the budget. If bits are relaxed to be continuous and $d_b \propto 4^{-b}$, the stationary rule becomes

$$b_i^* = \log_2 a_i + \log_2 \|v_i - o\| + c. \quad (5)$$

Equation 5 explains the reverse-water-filling shape, but Eq. 4 is the optimizer used in experiments. Adding the value factor changes our aggregate ladder-width statistic by at most 0.035 bits in the measured configurations; this is not a per-token bound and does not replace a direct allocation ablation.

Two coordinates, not one universal phase boundary. If $d_b \geq d_0$, tier b consumes bits without lowering the surrogate cost and is therefore dominated. This condition is sufficient for extinction; $d_b < d_0$ is not sufficient for membership in the lower convex envelope. We summarize the sharp side by

$$D_2 = H^{-1} \sum_{h=1}^H \mathbb{I}[d_{h,2} \geq d_0], \quad (6)$$

the fraction of heads in a model–context cell whose 2-bit tier is dominated. The diffuse side is different: unequal allocation has little leverage when the spread of token importance is narrower than the tier spacing. For attention-only importance, $\text{std}_i(\log_2 a_i) = \tau / \ln 2$ is an algebraic identity, not evidence of optimality. Thus D_2 primarily measures approach to the eviction-like corner; a dispersion coordinate is also required to establish the uniform corner. Finally, D_2 contains no explicit L , but it is measured *at* a context and changes with context.

3 MEASUREMENT PROTOCOL

Models, contexts, and statistical units. We evaluate Llama-3.1-8B and Llama-3.3-70B (Grattafiori et al., 2024), Mistral-7B (Jiang et al., 2023), Qwen1.5-MoE-A2.7B (Qwen Team, 2024), Qwen3-8B, and Qwen3-30B-A3B (Yang et al., 2025). The strict symmetric R3 grid has 16 model–context cells between 8k and 128k and 29,312 query-head–configuration aggregates. Boundary extension R6 has 25 cells spanning 4k–256k and 44,416 such aggregates. Across the six architectures there are 10,240 query-head positions and 2,016 KV-head positions counted once; contexts reuse those heads. Statistical claims therefore treat models/cells, not raw heads, as independent units. PG-19 text (Rae et al., 2020) supplies the haystack, and every retrieval cell passes a full-cache needle-retrieval gate.

Compression and budget. Keys are quantized with TurboQuant_{mse} (Zandieh et al., 2025), using random rotation and Lloyd–Max scalar quantization (Lloyd, 1982). The headline budget is $B = 3$ *key bits per context token*; tier 0 evicts and tier 8 stores an 8-bit key. Values remain uncompressed, and the accounting excludes scale, index, metadata, protected-window, and packing costs. It is therefore a matched key-bit experiment, not a matched total-KV-byte system comparison.

Information contract and endpoints. The primary R3/R6 comparison is symmetric: both the mixed allocator and eviction endpoint use only lagged attention information, while positions unseen by that score remain at the maximum tier. The endpoints are uniform 3-bit key quantization and accumulated-attention eviction retaining the number of 8-bit keys allowed by the same key-bit budget. The headline corner set is named *uniform+accum*; a robustness analysis takes the minimum error over five available corners. Oracle/current-query results are retained only as upper bounds. The original allocation is per query head; the co-design study separately enforces one allocation per shared KV head.

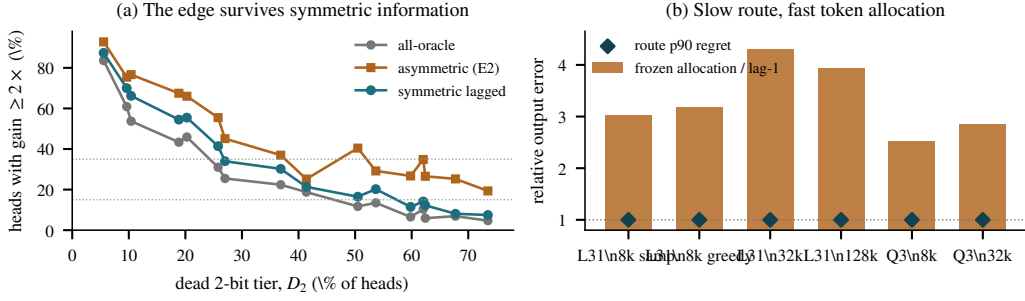


Figure 2: **Measured robustness and timing.** (a) On the same 16 cells, the honest lagged-versus-lagged advantage is smaller than the asymmetric E2 estimate but remains ordered by D_2 . The band is secondary and depends on the named corner set. (b) In six R5 cells, retaining the calibration-time *policy choice* has p90 regret 1.00 through 4,096 generated tokens, whereas freezing the token assignment costs $2.52\text{--}4.30\times$ a one-step-old assignment at the valid horizon.

Exact response and metrics. For every assignment we quantize the real keys, recompute attention, and measure the resulting output error exactly. For head h ,

$$g_h = \frac{\min(D_h^{\text{uniform}}, D_h^{\text{evict}})}{D_h^{\text{mixed}}}, \quad G_+ = \exp \left[H^{-1} \sum_h \log \max(g_h, 1) \right]. \quad (7)$$

G_+ is the primary continuous cell response. It is a retrospective, hindsight portfolio statistic, not achieved router performance. The share of heads with $g_h \geq 2$ (the “band”) is a secondary operational summary. Correlations are reported over cells, with model-partialled and model-resampled checks where available.

4 WHEN DOES THE INTERIOR PAY?

The advantage survives a fair information comparison. In the 16-cell R3 grid, demoting both the mixed allocator and eviction endpoint to lagged information retains roughly half of the apparent asymmetric advantage. The symmetric band remains 1.2–12.5 points above the all-oracle band in all 16 cells, but five cells are now STOP (band < 15%) and five are NARROW (15–35%). The continuous hindsight portfolio gain ranges from $1.05\times$ to $3.21\times$. Most importantly for the regime claim, D_2 still orders the symmetric band at $\rho = -0.985$. Figure 2a exposes all three information contracts rather than labeling the asymmetric E2 result as symmetric.

Tier extinction orders value, but the crossing is model-conditioned. Extending to 25 cells, D_2 has Spearman $\rho = -0.978$ with both G_+ and the band; residualizing model identity leaves $\rho = -0.915$. A log-linear fit has leave-one-model-out error of 2.60 band points. Pooled monotone crossings occur near $D_2 = 29.0\%$ (GO) and 54.1% (STOP), but they are not universal thresholds: Llama-3.1-8B remains at 20.2% band at $D_2 = 53.7\%$, while Qwen3-8B is at 12.2% band at $D_2 = 54.3\%$. This eight-point disagreement across 0.6 points of D_2 exceeds prompt uncertainty and identifies architecture scatter. We therefore use D_2 as an ordering coordinate and fit any operational boundary per architecture.

The band also saturates near zero and can exhibit noise-level inversions, whereas G_+ remains monotone along the measured Qwen3-8B trajectory. This is why Fig. 1a and our claims lead with the continuous response. Prompt-block standard deviation is 0.4–3.0 band points, while replacing the accumulated-attention corner with the five-corner minimum shifts the band by -5.1 to -8.6 points and can change a verdict. Every band number must therefore name its information and corner sets.

TBD [R10: predictor comparison] Compare D_2 prospectively with RateQuant’s AM/GM or $\text{var}(\log w)$ heterogeneity, entropy/Gini, and context-only baselines under a fixed leave-model-out protocol.

Table 1: Completed deployment-facing measurements. These are output-error results, not end-task or throughput claims.

Constraint	Measurement	Implication
Family label over 4,096 tokens	p90 output-error regret 1.00 in 6/6 cells	Temporally stable in these measured cells
Frozen token assignment	2.52–4.30× lag-1 error	Re-budget mixed heads quickly
Shared GQA allocation	1.000/1.136/1.316× at $n_{\text{rep}} = 1/4/8$	Route per KV head, with a measured discount
4-bit current-query cascade	closes 82–91% (median 89%) of lag penalty	Recover scores from base-tier keys

Context traverses the map. Every measured model trajectory moves generally toward larger tier extinction with context, though no single global boundary fits all architectures. Llama-3.3-70B moves from GO at 8k to NARROW at 128k; Qwen3-8B moves from NARROW at 8k to STOP by 16k under the uniform+accum corner. The mechanism is mostly an absolute-length effect: τ is generally convex in $\log L$. Reaching the trained/RoPE cap adds a real residual of about +0.21 in Qwen3-8B and Qwen3-30B, but no comparable excess in Llama-3.1-8B and none at 0.75 of the cap. Context and architecture therefore both matter; the data do not support a universal RoPE-triggered transition.

The surrogate is useful, not exact. Across the original configuration grid, the median ratio between surrogate-predicted and exactly recomputed gain lies in $[0.79, 1.34]$. Because tier costs and evaluation data are not yet fully disjoint, we treat this as model-fit validation rather than held-out prediction. Exact recomputation protects the empirical ordering from the approximation, but it does not validate every term of Eq. 2.

TBD [A-VALUE] Run a direct value-aware versus attention-only allocation ablation, reporting tier-assignment changes and exact output error; the current 0.035-bit result concerns only an aggregate width statistic.

5 DEPLOYMENT IMPLICATIONS AND LIMITS

The completed co-design experiments expose two refresh timescales in attention-output error. A head’s output-error-optimal family label was stable over 4,096 generated tokens in six measured cells, while token assignments inside a mixed head required a short refresh schedule (Fig. 2b). This stability does not establish an end-task router. The relevant measured costs are:

These facts motivated the tested SIEVE selector: estimate the regime at a deployment context, choose a uniform, mixed, or eviction-like policy per KV head, and refresh mixed-head assignments from a 4-bit base pass. Under the completed end-task contract, however, the calibrated selector is not a safe general policy. The design remains an output-error hypothesis rather than an achieved end-task method.

End-task transfer. The completed R8 campaign does not validate the output-error map as an accuracy router. Uniform quantization is the strongest fixed arm in 34/36 valid task–budget cells; the calibrated router has 17 clear losses, all on Llama, and the phase-band/accuracy-gain Spearman correlation is -0.70 , opposite the prespecified direction. The new sparse baselines improve on SnapKV but do not reverse uniform’s broad advantage. On two natural forced-choice development blocks, the label-seeing candidate oracle adds only 3/52 over the best fixed Llama policy and a descriptive 2/52 for Qwen; the Qwen protocol formally stops first because FP accuracy is 24/52 < 0.50. Both proxy outcomes are suppressed by sequential gates, and confirmation remains unopened.

TBD [SYSTEM] Measure nested-tier storage, scale/index/metadata bytes, protected-window cost, and packed-kernel latency/throughput. The current experiment establishes key-bit output error, not total-KV memory or speedup.

Table 2: Closest work and the claim boundary of this paper.

Work	Allocation/action	Difference here
DiffKV; HqKV	token high/low precision and pruning	explain and predict when the middle tier pays; no first-hybrid claim
CAOTE; ReST-KV	output-aware token eviction	finite-rate extension under a stated local surrogate; exact set error retained
RateQuant	K/V heads, finite rates	within-head key tokens, explicit zero rate, context traversal
RDKV (concurrent)	value tokens, key channels, zero-to-finite rates	when the token-wise interior beats endpoints; score staleness and regime movement
AATC (concurrent)	transformed K/V channels, including zero	token-axis tier viability; values/channels remain outside our allocation

6 RELATED AND CONCURRENT WORK

Quantization, eviction, and hybrid token storage. KV quantizers use per-channel/per-token integer schemes (Liu et al., 2024; Hooper et al., 2024), rotations or sketches (Zandieh et al., 2025; 2024), and salient-token mixed precision (Yang et al., 2024; He et al., 2024). Evictors use accumulated or windowed attention (Zhang et al., 2023; Li et al., 2024; Liu et al., 2023; Xiao et al., 2024) and allocate keep budgets across layers or heads (Cai et al., 2024; Feng et al., 2025). The historical split is not strict: archival DiffKV routes tokens among high precision, low precision, and pruning and supplies an on-GPU manager (Zhang et al., 2025); HqKV optimizes among full precision, several quantized levels, and eviction using joint K–V importance (Wang et al., 2026). Our novelty is not hybrid tiering. We derive and test when an intermediate key-token tier improves over both endpoints.

Output-aware eviction. CAOTE derives the exact singleton attention-output eviction score used in Eq. 3 (Goel et al., 2025). ReST-KV models attention redistribution through layer-wise output reconstruction and spatial-temporal smoothing (An et al., 2026). We do not claim first output-aware eviction; we state the separable approximation needed to include zero rate in a finite-rate allocator and study the regimes that result.

Rate allocation along complementary axes. RateQuant fits quantizer-specific distortion curves and allocates finite precision across K/V heads, with a heterogeneity predictor for mixed-precision gain (Zuo et al., 2026). Concurrent RDKV assigns zero-to-finite rates to value tokens and key channels and implements packed decode (Zhang et al., 2026). Concurrent attention-aware transform coding (AATC) applies output-aware reverse water-filling to transformed K/V channels, including zero-bit rank reduction (Laus et al., 2026). Our allocation unit is instead the within-head *key-token*; our target is the context-dependent value of the interior, not the existence of water-filling itself. Head, channel, and token axes are potentially composable.

7 LIMITATIONS AND CONCLUSION

The current evidence is deliberately scoped. The completed R8/P2 campaign shows that attention-output error does not provide a validated task-accuracy router under the evaluated contract. The allocation is key-token only, leaves values exact, and does not match total KV bytes or system overhead. The regime crossing is architecture-conditioned, and the diffuse/uniform side has less empirical support than tier extinction. The within-generation change of τ is underpowered at 3–6 prompts per cell and needs roughly 20; it is not used for the two-timescale conclusion. Finally, a packed implementation is required to translate key-bit savings into throughput.

Within that scope, the conclusion is stable: joint allocation is not uniformly useful. A tier-extinction coordinate strongly orders when its interior lowers exact attention-output error relative to uniform quantization and eviction, context can move a fixed model toward a corner, and the slow-family/fast-allocation split identifies two refresh costs that any future output-error controller must address. The evaluated task results set a separate boundary: this local ordering cannot be presented as downstream effectiveness.

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CONTENTS

1	Introduction	1
2	Joint token-rate allocation as a common foundation	2
3	Measurement protocol	3
4	When does the interior pay?	4
5	Deployment implications and limits	5
6	Related and concurrent work	6
7	Limitations and conclusion	6
A	Some Basics	9
B	Scope of the common rate-allocation view	10
C	Output distortion and the zero-rate approximation	10
C.1	Attention and finite-bit key perturbations	10
C.2	Exact set eviction	11
C.3	The planning approximation	11
D	Discrete rate allocation and regime coordinates	11
D.1	The implemented optimizer	11
D.2	Tier extinction	12
D.3	Why the regime map needs two coordinates	12
E	Measurement protocol and accounting	12
E.1	Grid and independent units	12
E.2	Matched information and budget	13
E.3	Responses	13
F	Additional robustness and deployment evidence	13
G	Claim boundary	13

A SOME BASICS

The tasks are synthetic retrieval (RULER): find a random number planted somewhere in 8K–128K tokens of text, with distractor numbers nearby. There are 20 prompts per cell.

- Synthetic Retrieval: This means the test is artificially constructed rather than being a natural real-world task (like summarizing a real book). A piece of data is intentionally hidden inside a massive wall of text just to see if the AI can find it.
- RULER: This is the specific name of the benchmark framework (developed by researchers to evaluate long-context LLMs). It stands for Retrieval, Understanding, and Long-context Evaluation Robustness.
- 8K-128K tokens: This indicates the length of the text context that the model has to process. A token is roughly a word or part of a word, so 8K tokens is about 8,000 words, and 128K tokens is about 128,000 words. The model has to search through this large amount of text to find the hidden number.
- Distractor numbers: These are other numbers placed in the text that are not the target number. They are meant to confuse the model and make the task harder. The model has to correctly identify the planted number and ignore the distractors.
- 20 prompts per cell: The Matrix (Cells): When testing LLMs, researchers usually create a grid or matrix. For example, the horizontal axis might be the context length (8K, 16K, 32K, 64K, 128K) and the vertical axis might be the depth of where the needle is hidden (0% depth = top of the document, 50% = middle, 100% = bottom). Each intersection on this grid is a "cell." This means that for each configuration or "cell" of the experiment (which could be a specific model, context length, or other parameter), there are 20 different test cases (prompts) that the model is evaluated on. Each prompt is a separate instance of the retrieval task.

Some metrics

- valid mean task score

10 prompts -> 100 prompts or 1000 prompts? not head-output error but task score. The task score is the final evaluation metric that measures how well the model performed on the retrieval task. It is calculated based on whether the model correctly identified the planted number in each prompt. The valid mean task score is the average score across all valid prompts (those that were not filtered out due to errors or other issues). This metric gives an overall sense of the model's performance on the retrieval task across different configurations and contexts.

B SCOPE OF THE COMMON RATE-ALLOCATION VIEW

This appendix expands the derivation and measurement contract used in the main paper. The common action space is deliberately narrow: it allocates *key-token* rates within an attention head, leaves values exact, and measures attention-output error. Prior work already combines quantization and eviction in deployable systems. Our claim is instead that a common local surrogate makes it possible to ask a different empirical question: when does the discrete interior improve over uniform quantization and eviction under matched information and key-bit budgets?

Throughout, *zero rate* means a zero-bit action in the separable planning surrogate. It does not mean that set eviction is exactly identical to independent additive logit noise. Section C makes this distinction explicit.

C OUTPUT DISTORTION AND THE ZERO-RATE APPROXIMATION

C.1 ATTENTION AND FINITE-BIT KEY PERTURBATIONS

For a query $q \in \mathbb{R}^d$, keys $k_i \in \mathbb{R}^d$, and values $v_i \in \mathbb{R}^{d_v}$, define

$$s_i = q^\top k_i / \sqrt{d}, \quad a_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)}, \quad o = \sum_i a_i v_i. \quad (8)$$

Quantizing key i changes its logit by δ_i . Because

$$\frac{\partial o}{\partial s_i} = a_i(v_i - o),$$

the first-order output perturbation is

$$\Delta o \approx \sum_i a_i (v_i - o) \delta_i. \quad (9)$$

Consequently,

$$\begin{aligned} \mathbb{E} \|\Delta o\|^2 &\approx \sum_i a_i^2 \|v_i - o\|^2 \text{Var}(\delta_i) \\ &\quad + \sum_{i \neq j} a_i a_j (v_i - o)^\top (v_j - o) \text{Cov}(\delta_i, \delta_j). \end{aligned} \quad (10)$$

Equation 2 follows only when the residual perturbations are zero mean and the cross-token covariance contribution vanishes. Softmax invariance removes a *common* shift, $\text{softmax}(s + c\mathbf{1}) = \text{softmax}(s)$; it does not remove arbitrary token-dependent bias.

C.2 EXACT SET EVICTION

Let E be an evicted token set and $A_E = \sum_{i \in E} a_i$. Renormalizing the remaining attention weights yields

$$o_{\setminus E} = \frac{o - \sum_{i \in E} a_i v_i}{1 - A_E}, \quad (11)$$

$$o_{\setminus E} - o = -\frac{\sum_{i \in E} a_i (v_i - o)}{1 - A_E}, \quad (12)$$

$$\|o_{\setminus E} - o\|^2 = \frac{\|\sum_{i \in E} a_i (v_i - o)\|^2}{(1 - A_E)^2}. \quad (13)$$

The denominator and cross-token inner products prevent an exact token-separable representation. For one evicted token j , the exact error magnitude is

$$\frac{a_j}{1 - a_j} \|v_j - o\|,$$

which connects directly to the CAOTE singleton score. ReST-KV further models attention redistribution and output reconstruction; our local planner does not subsume those effects.

C.3 THE PLANNING APPROXIMATION

When the evicted mass is small and cross terms are neglected, Eq. equation 13 motivates

$$\tilde{D}_0(E) = \sum_{i \in E} w_i, \quad w_i = a_i^2 \|v_i - o\|^2.$$

We choose units in which the absolute zero-rate distortion is $d_0 = 1$ and write the finite-rate costs as $d_b = \text{Var}(\delta_b)$ in the same logit units. Thus

$$\tilde{D}(\mathbf{b}) = \sum_i w_i d_{b_i}, \quad b_i \in \mathcal{B}, \quad 0 \in \mathcal{B}.$$

This convention keeps relative quantizer factors separate from the absolute zero-rate cost. It is a local approximation used to choose an allocation; every reported response is obtained by recomputing attention with the actual quantized/evicted cache.

D DISCRETE RATE ALLOCATION AND REGIME COORDINATES

D.1 THE IMPLEMENTED OPTIMIZER

At average key budget B , the discrete problem is

$$\min_{b_i \in \mathcal{B}} \sum_i w_i d_{b_i} \quad \text{subject to} \quad \sum_i b_i \leq BL. \quad (14)$$

Introducing multiplier $\lambda \geq 0$ separates the token decisions:

$$b_i^*(\lambda) = \arg \min_{b \in \mathcal{B}} \{w_i d_b + \lambda b\}. \quad (15)$$

We search λ to satisfy the budget and use the discrete tiers in Eq. equation 15; the continuous expression is explanatory, not the experimental optimizer.

For the relaxation $d_b = \kappa 4^{-b}$,

$$b_i^* = \frac{1}{2} \log_2 w_i + c = \log_2 a_i + \log_2 \|v_i - o\| + c.$$

The measured 0.035-bit result is the maximum observed change in an *aggregate ladder-width statistic* after adding the value term. It is not a uniform per-token bound and does not establish that attention-only allocation is optimal. A direct value-aware versus attention-only allocation ablation is therefore retained as an explicit evidence anchor.

D.2 TIER EXTINCTION

If $d_b \geq d_0$, tier b consumes positive bits without improving the local surrogate relative to zero rate and is dominated. This is a sufficient extinction condition. The converse is not sufficient: a tier with $d_b < d_0$ can still lie above the lower convex envelope of the other rate–distortion points. For the 2-bit tier we report

$$D_2 = \frac{1}{H} \sum_{h=1}^H \mathbb{I}[d_{h,2} \geq d_0].$$

D_2 is measured at a particular model and context. Although L is absent from its formula, the head cost distribution changes with L , so D_2 is not sequence-length invariant and a new deployment context requires calibration.

D.3 WHY THE REGIME MAP NEEDS TWO COORDINATES

The sharp side and diffuse side arise for different reasons:

1. On the sharp side, large absolute finite-bit noise removes low-bit tiers; D_2 directly probes this boundary.
2. On the diffuse side, unequal allocation has little leverage when token importances are nearly equal; an importance-dispersion statistic is needed.

For attention-only importance, $\text{std}_i(\log_2 a_i) = \tau / \ln 2$, because softmax subtracts the same log normalizer from every logit. This identity alone neither proves that τ is a complete regime variable nor establishes a uniform-quantization optimum. The paper therefore treats the mixed-to-eviction side as the well-supported boundary and leaves a prospective comparison of dispersion predictors as R10.

E MEASUREMENT PROTOCOL AND ACCOUNTING

E.1 GRID AND INDEPENDENT UNITS

The symmetric R3 grid contains 16 model–context cells from six models and 29,312 query-head–configuration aggregates. The R6 boundary extension contains 25 cells spanning 4k–256k and 44,416 such aggregates. Across the six models, 10,240 query-head positions and 2,016 KV-head positions are counted once; contexts reuse those positions. Model/cell-level analyses, rather than raw heads, are therefore the independent statistical units.

PG-19 passages supply natural context. Each cell is admitted only after the uncompressed model passes the needle-retrieval validity gate. The allocation planner uses its local surrogate, while every reported error comes from actual key quantization or eviction followed by exact attention-output recomputation.

E.2 MATCHED INFORMATION AND BUDGET

The primary comparison uses $B = 3$ key bits per context token. Uniform quantization stores every key at 3 bits. The eviction endpoint retains the number of 8-bit keys allowed by the same key-bit budget. In the symmetric contract, the mixed allocator and accumulated-attention eviction both use lagged importance information; an oracle/current-query result is labeled only as an upper bound. The five-corner robustness result changes the reference set and is reported separately.

This accounting omits value-cache bytes, quantization scales, indices, metadata, protected windows, alignment, and packing. Moreover, *8-bit* is the maximum experimental key tier, not full floating-point precision. The experiments therefore establish matched key-bit output error, not matched total memory or throughput.

E.3 RESPONSES

For each head,

$$g_h = \frac{\min(D_h^{\text{uniform}}, D_h^{\text{evict}})}{D_h^{\text{mixed}}}, \quad G_+ = \exp\left(\frac{1}{H} \sum_h \log \max(g_h, 1)\right).$$

G_+ is a hindsight portfolio statistic: it clips losses after observing all three errors. It is not the performance of a deployed router. The primary cell-level response is continuous G_+ ; the fraction with $g_h \geq 2$ is a secondary thresholded band statistic. Every band result names its corner and information contracts.

F ADDITIONAL ROBUSTNESS AND DEPLOYMENT EVIDENCE

The completed measurements support the following bounded conclusions:

- In all 16 R3 cells, the symmetric lagged-versus-lagged band exceeds the all-oracle band by 1.2–12.5 points, but it is lower than the asymmetric current-query-versus-lagged estimate. Five cells are STOP and five NARROW.
- Across the 25 R6 cells, D_2 has Spearman correlation -0.978 with both G_+ and the band; the model-partial correlation is -0.915 . Similar D_2 values can nevertheless differ by eight band points across architectures.
- Prompt-block standard deviation is 0.4–3.0 band points. Replacing the two headline corners by the five-corner minimum lowers the band by 5.1–8.6 points and can change a categorical verdict.
- In six R5 cells, the calibration-time family choice has p90 regret 1.00 through 4,096 generated tokens, whereas a frozen token allocation costs $2.52\text{--}4.30\times$ a one-step-old allocation.
- Enforcing one allocation per shared KV head gives marginal in-band costs 1.000, 1.136, and 1.316 at GQA repetition factors 1, 4, and 8. A 4-bit current-query cascade closes 82–91% (median 89%) of the lagged-score penalty.

One predictor study remains intentionally unfilled. R10 must compare D_2 prospectively against RateQuant-style heterogeneity, entropy/Gini, and context-only predictors under a fixed leave-model-out protocol. R8/P2 and the four-baseline extension are complete with disjoint routes and reader/validity/false-positive/bit-accounting audits. Uniform is the strongest fixed arm in 34/36 valid cells, while the calibrated router has 17 clear losses. Two natural-task development blocks expose only 3/52 and 2/52 candidate headroom over their best fixed policies; sequential gates suppress both proxy outcomes and leave confirmation unopened. These results do not establish end-task effectiveness.

G CLAIM BOUNDARY

Supported	Not yet supported
Zero rate is useful inside a stated separable local surrogate	Exact equivalence between independent quantization noise and set eviction
An absolute tier cost can make a finite-bit tier dominated	A universal one-dimensional phase transition or threshold
D_2 strongly orders the measured mixed-to-eviction boundary	A context-free predictor that transfers without calibration
The output-error-optimal family label is temporally stable in six measured cells while token allocations become stale quickly	A robust end-task router, total-KV memory savings, or realized throughput

Table 3: The empirical and methodological claim boundary of the current draft.